Fiscal Year:	FY 2007	**Task Last Updated:**	FY 01/31/2008
PI Name:	Maurer, Richard H. Ph.D.		
Project Title:	Combined Ion and Neutron Spectrometer for Space Applications		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Technology Development Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	richard.maurer@jhuapl.edu	**Fax:**	FY 240-228-6696
PI Organization Type:	UNIVERSITY	**Phone:**	240-228-6482
Organization Name:	The Johns Hopkins University		
PI Address 1:	Applied Physics Laboratory		
PI Address 2:	11100 Johns Hopkins Road		
PI Web Page:			
City:	Laurel	**State:**	MD
Zip Code:	20723-6099	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	08/01/2004	**End Date:**	08/31/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Goldsten, John (The Johns Hopkins University Applied Physics Laboratory) Roth, David (The Johns Hopkins University Applied Physics Laboratory) Zeitlin, Cary (Lawrence Berkeley National Laboratory) Haggerty, Dennis (The Johns Hopkins University Applied Physics Laboratory)		
Grant/Contract No.:	NCC 9-58-TD00404		
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Task Description:	We proposed to design and fabricate a Combined Ion and Neutron Spectrometer (CINS) for Space Applications. This instrument will improve upon existing charged particle species and energy spectrometers (e.g., MARIE on Mars Odyssey) and incorporate the capability of the previously developed NSBRI Neutron Energy Spectrometer to yield a single and complete ionizing radiation environment monitor for application in space habitats and transport vehicles. CINS will be used to compare its energy spectra with the Tissue Equivalent Proportional Counter (TEPC) to investigate basic hypotheses about the lack of radiation environment information from the TEPC LET (Linear Energy Transfer) spectrometer due to detection limitations with respect to high energy protons, low energy heavy ions and neutrons. The originally proposed objectives were refined with respect to specific metrics in February/March 2005 to 1) eliminate the gain saturation of MARIE for heavy ions with LET > 35 keV/micron; 2) increase the dynamic range vs. the MARIE instrument by a factor of 100-200 to a minimum of 10,000:1 to include protons with energies above 100 MeV; 3) increase the maximum instrument event detection rate by at least a factor of 10 to a minimum of 100 Hz. The end product of the effort will be a Countermeasure Readiness Level 6 instrument. The conceptual design of the CINS instrument was facilitated through the use of Monte Carlo simulations that model the basic detector stack, establish the energy resolution, and quantify the issues that will affect the instrument's efficiency using GEANT4. Our CINS design consists of a stack of 7 cylindrical detectors with a radius of 20 mm. There are 3 different types of detectors in the CINS stack. One is a standard Silicon (Si)solid state detector. The remaining two detector types are scintillators. The first detector in the stack is a 1 mm thick BC430 detector followed by a 5 mm thick Si detector, then a second 1 mm BC430 detector, followed by three 5 mm Si detectors. The final detector is a 30 mm thick detector of BGO. The entire stack will be wrapped in a 3mm thick Titanium shell. For the CINS system to accurately resolve solar energetic particle (SEP) events the system must cleanly separate the protons from the electrons. We simulated spectra of electrons to 20 MeV and protons to 1 GeV. While a small percentage of the electrons and protons do overlap in this simulation, the vast majority of the protons can be separated from the electrons. Protons up to ~150 MeV are stopped in the stack while protons with E>150 MeV penetrate through the system. Given reasonable energy resolution of the BGO this simulation shows that protons up to ~300 MeV can be uniquely identified. Helium spectra up to 750 MeV/nucleon and O, Si, and Fe spectra well past their GCR maxima (1 GeV/nucleon) are cleanly resolved. We re-drifted eight Si(Li) detectors for charged particles and neutrons from LBNL's existing stock. The second step was the creation of a guard ring accomplished by routing out the detector in a circular pattern at a fixed radius. These eight detectors have been evaluated in experiments at HIMAC and NSRL during 2006 and 2007. For the charged particle telescope we procured a 38mm X 38mm BGO scintillator detector unit and obtained 1mm thick scintillators. We designed and fabricated detector boards for NSRL evaluations and designed charged particle telescope electronics for control, power and data acquisition. We are presently developing the back end electronics including the digital subsystem. Experiments were executed at NSRL in March 2006 by colliding a 1 GeV/amu iron beam with a 5 inch thick aluminum target simulating a large space vehcle. The progeny of these collisions, both charged particle fragments and neutrons, were monitored by the refurbished 5mm thick silicon detectors. During May 2007 five of the seven detector elements of the charged particle telescope were operating and a large volume of charged particle fragment data was obtained. For the 2007 experiments targets of 10-20 g/cm^2 were used. For the Neutron Energy Spectrometer JHU/APL has designed, procured and fabricated a 12cm X 12cm Eljen plastic boron-loaded scintillator detector for medium energy 1-14 MeV neutrons that have the maximum dose equivalent weighting factors of 10-20. This detector system was calibrated at Columbia University RARAF in November 2006 and used at NSRL for the first time in May 2007. Schedule The overall schedule from the proposal can be divided into three phases of duration of 16 months each-- August 04-November 05--instrument design including modeling and fabrication December 05-March 07--finish fabrication, calibration and evaluation of detector systems; detector technology experiments at NSRL April 07-July 08--accelerator experiments with spacecraft shielding configurations and comparison with TEPC LET spectrometer for the charged particle and neutron environment. Schedule Details 2006: A) Procure and assemble the BGO scintillator for the charged particle telescope; B) Assemble the Eljen medium energy neutron detector system; C) Perform medium energy neutron detector evaluation and calibration at Columbia RARAF; D) Publish results of the March 2006 NSRL thick silicon detector experiments. 2007: A) BGO scintillator evaluation experiments at HIMAC and NSRL (Spring 07); B) Medium energy neutron collision experiments at NSRL (Spring 07); C) First experiments with the assembled charged particle telescope at NSRL (Summer 07). Fall 2007/2008: A) Continue target collision experiments with all 7 detector elements of the charged particle telescope operating and with data acquired with the new back end electronics; B) Repeat experiments with the Eljen scintillator for neutron detection employing a better signal to noise ratio for the NSRL target room; C) compare results with those from a TEPC LET spectrometer.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Neutron detection for homeland security is of interest in the areas of 1) the location of radiological weapons used by terrorists, 2) scanning and inspecting imported cargo in ship holds or on the dock side and 3) smart portals for the scanning of personnel or small packages at selected points of entry.

Task Progress:	Charged particle telescope development over the last funded year: We re-drifted eight Si(Li) detectors for charged particles and neutrons from LBNL's existing stock. The second step was the creation of a guard ring accomplished by routing out the detector in a circular pattern at a fixed radius. These eight detectors have been evaluated in experiments at HIMAC and NSRL during 2006 and 2007. For the charged particle telescope we procured a 38mm X 38mm BGO scintillator detector unit and obtained 1mm thick scintillators. We designed and fabricated detector boards for NSRL evaluations and designed charged particle telescope electronics for control, power and data acquisition. We are presently developing the back end electronics including the digital subsystem. This includes the use of an FPGA (Field Programmable Gate Array). During May 2007 five of the seven detector elements of the charged particle telescope were operating and a large volume of charged particle fragment data was obtained. For the 2007 experiments targets of 10-20 g/cm^2 were used. Neutron Energy Spectrometer: For the Neutron Energy Spectrometer JHU/APL has designed, procured and fabricated a 12cm X 12cm Eljen plastic boron-loaded scintillator detector for medium energy 1-14 MeV neutrons that have the maximum dose equivalent weighting factors of 10-20. This detector system was calibrated at Columbia University RARAF in November 2006 and used at NSRL for the first time in May 2007.
Bibliography Type:	Description: (Last Updated: 04/27/2011)
Articles in Peer-reviewed Journals	Maurer RH, Roth DR, Kinnison JD, Haggerty DK, Goldsten JO. "The NSBRI/APL neutron energy spectrometer." Johns Hopkins APL Technical Digest. 2006 Jan; 27(1):56-65. , Jan-2006